

Department of Mechanical and Mechatronics Engineering



ME 380 - Mechanical Engineering Design Workshop

Final Design Report

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Executive Summary

The “Gorilla Gripper” design project involved designing, building, testing, and verifying the performance of a pick and place manipulator to move objects between specified pick-up and drop-off areas. The primary design objectives selected are strength to weight ratio to lift and move heavy objects and accuracy/precision for pick and place repeatability. Consequently, the needs statement is as follows; “A need exists to design a system with a strength to weight ratio of at least 1:1 that can precisely move a benchie across a 2D surface and place it in a container with 2 inch high walls.”

This was addressed through developing, integrating, and implementing a gripper, boom, and rotating base assembly. Firstly, the gripper design involves 2 gripper arms that are actuated using a stepper motor. This is done through translating rotational motion from the stepper motor to linear motion using a threaded rod in combination with a sled attached to the arms. This allows the gripper arms to effectively open and close to pick and place the object. The parts involved in the final gripper design include a stepper mount, shaft couple, sled, gripper arms, a threaded rod, a bushing for the rod, and a housing for that bushing.

Additionally, the boom sub-assembly involves a chassis that is integrated with the linear actuator allowing the gripper device to move vertically. The chassis supports several components including the gripper, linear rail interface, linear actuator, etc. Lastly, the rotating base sub-assembly consisted of the bearing housing, the rotating shaft, the mounting plate, and both the ring and pinion gears. This device mounted the manipulator to the base structure and allowed the gripper and boom to rotate around a fixed axis.

The engineering design specifications were verified and tested with several test cases. To test the strength of the system, different loads were applied to the parts during the prototyping phase. The lifting capacity was determined through incrementally increasing the weight of the object being lifted until the system failed. To test the precision/accuracy objective a 3 inch diameter hockey puck was repeatedly picked and placed within specified circular locations. The error of the system was found by recording the distance from the edge of pick up/ drop off location to the edge of the hockey puck.

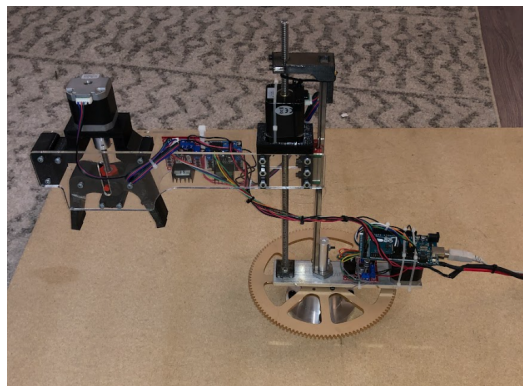


Figure 1: Pick and place design space requirement

1.0 Introduction

1.1 Project Background

The design project requirements for ME 380 involved designing, building, testing, and verifying the performance of a pick and place manipulator to move objects between specified pick-up and drop-off areas. Additionally the following project constraints were adhered to during the initial design selection phase (see fig. 2):

- The red and blue bins (pick up and drop off locations) should have at least 2" tall sides.
- The pick and place manipulator must be designed with at least one fixed base location between the pick-up and drop-off locations.
- Energy can be supplied to the system from one location in the user console area.
- Users cannot touch or make direct physical contact with the objects to assist with moving.
- Pick-up/drop-off areas cannot be manipulated to assist with object pick-up or placement.
- No mobile/flying/launching/throwing manipulator designs.

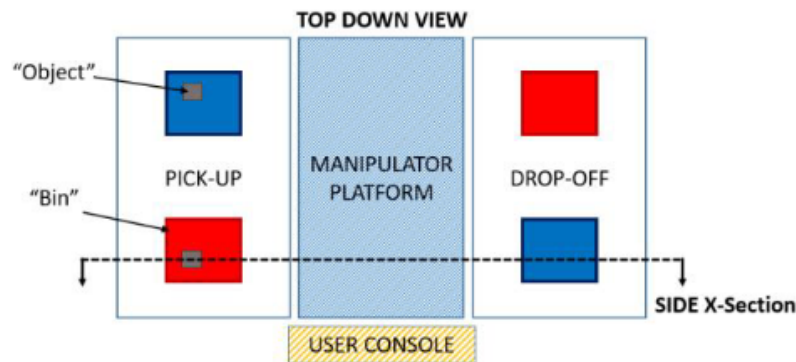


Figure 2: Pick and place design space requirement

The project scope was split into two phases; designing and building a gripper device, and designing a manipulator to move the object. In response to these requirements, the design objectives of optimizing strength to weight ratio and accuracy/precision for pick and place repeatability were selected.

1.2 Needs Analysis

The needs statement for the project is as follows; “A need exists to design a system with a strength to weight ratio of at least 1:1 that can precisely move a benchie across a 2D surface and place it in a container with 2 inch high walls.”

Based on the selected design objectives and the project scope, the following constraints and criteria were outlined during the initial design selection:

Table 1: Project Constraints and Criteria

Criteria	Constraints
● Maximize strength-to-weight ratio ● Maximize placement accuracy ● Minimize cost	● Benchy lifted at least 2 inches vertically ● Translate at least 12 inches diagonally ● Cost less than \$300 CAD ● No physical interaction with the system

Moreover, the system was designed with consideration of assembly line applications where a small scale autonomous crane could be used. Further design specifications were outlined based on the selected project objectives to optimize strength and accuracy of the manipulator.

Table 2: Engineering Design Specifications

<u>Design Specification</u>	<u>Justification</u>
● Successfully Complete Benchy Pick and Place Test ● Total Component Cost < \$300	Adhere to ME 380 Project Requirements (see Project Background)
1:1 Strength-Weight Ratio	Parts can change weight during assembly (through welding or other processes) and the gripper must be able to lift the parts as they get heavier.
Precise Placement with error < 2% of object radius	Assembly lines require precise positioning of objects for subsequent manufacturing steps. For small objects this is a very small error band (For a hockey puck /w radius 1.5" => error < 0.031" or 0.79mm).
Continuous Running Precision	Assembly lines require movement of multiple objects without powering down or recalibration

2.0 Design Implementation

2.1 Gripper design

The first step for the design of the gripper claw included the brainstorming of different methods for actuating the claw. For this, each member of the team was tasked with creating a concept sketch for a possible gripper design. The result of this can be seen in the figures below.



Figure 3: Rack and Pinion Claw
(Jensine Aure)

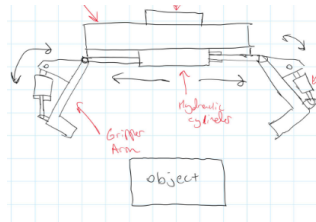


Figure 4: Hydraulic Claw
Concept (Salman Rasheed)

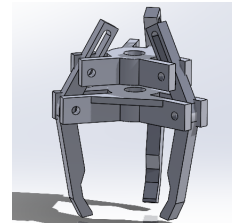


Figure 5: 3 Finger Claw (Tyson
Wagler)

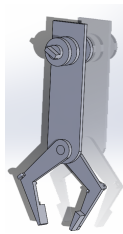


Figure 6: Spring Loaded Claw (Kwaku
Richter)

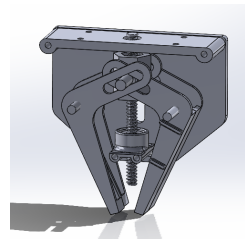


Figure 7: Power-Screw Actuated Claw
(Justin Siriska)

During a team meeting, the three concept ideas that appeared to be the easiest to implement and have both the required strength and accuracy for our main objectives were prototyped and further compared with the following decision matrix.

Table 3: Claw Decision Matrix

Design	Mechanical Reliability	Scalability to lift at least 15 lbs	Ease of Construction	Ease of Mounting to boom	Grip strength	Total
Figure 5	2	1	3	2	1	9
Figure 6	1	2	3	2	Couldn't Test	8
Figure 7	3	3	3	2	3	14

Overall, it was decided to use the power-screw actuated claw (Figure 7) because it proved to be the best in all the gripper specifications. In addition to the decision matrix, preliminary testing was performed by attaching a drill on a low torque setting to the prototype's threaded rod. The power-screw actuated prototype proved to have a very high grip strength which could easily lift 15 pounds. In addition to this, it had a simple and inexpensive construction as well as a uniform structure which would make it easy to integrate with the future boom design. During the initial testing it did not show signs of failure.

2.2 Rotating base, vertical lift, and boom

2.2.1 Original concept

After informal discussions on what the final design should look like, a rough concept sketch was drawn (shown in figure 8). Overall, the manipulator was to be composed of three subsystems: the horizontal boom, the vertical lift, and finally, the rotating base. The purpose of each system and how the design of each iterated throughout the design process is outlined in the subsequent sections below.

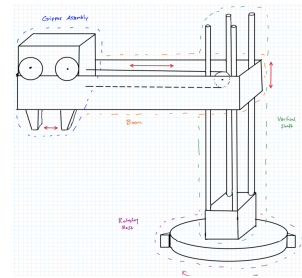


Figure 8: Concept sketch of overall manipulator design

2.2.2 Horizontal boom design iterations

The horizontal Boom was required to house the gripper assembly and provide the necessary translation of the object picked up when rotated by the base. The initial design included a horizontal positioner for the gripper to increase the range of reach of the manipulator. However, due to budgeting and time constraints, this was cut from the design as described in the design iterations below.

Iteration 1 - Boom with horizontal positioning of gripper

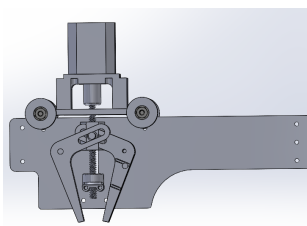


Figure 9: Horizontal boom with gripper positioner

Pros

- More degrees of freedom for gripper.

Cons

- Additional Stepper motor and H-bridge increase cost significantly.
- More weight and higher complexity due to extra moving parts.

By having the gripper linearly move in and out along the boom, the reachable range of the manipulator was greatly increased. However, due to the high cost of the electrical components, the design had to be simplified as detailed in iteration 2.

Iteration 2 - Boom with permanently fixed gripper

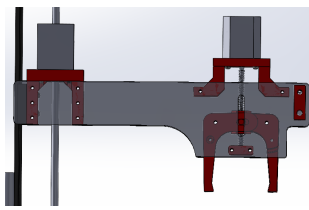


Figure 10: Boom with fixed mount gripper

Pros

- Simpler and cheaper design with less components.
- Gripper components directly integrated into boom strengthens both the boom and gripper assemblies.

Cons

- Less degrees of freedom of gripper (can't move in and out from crane).

With the gripper fixed in position, the range of the manipulator was restricted. However, because the gripper was mounted 10" from the center of rotation, the object could still be translated 20" with a 180° rotation of the base, which met the project requirements. The boom will be discussed in more detail in section 3.0.

2.2.3 Vertical lift design iterations

The vertical lift subsystem was required to position the horizontal boom accurately with at least 2" of vertical travel. To meet our main objectives, this was to be accomplished when the boom was carrying a load in the range of 0 to 15 lbs, in addition to the boom and gripper weight. Throughout the design process the vertical lift mechanism design was iterated based on analysis of the design at each step.

Iteration 1 - 4 post guide with centered linear actuator (Concept Sketch)

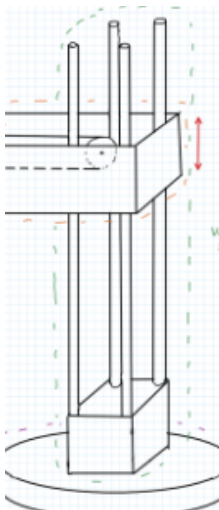


Figure : 11 Post linear guide

Pros

- Linear actuator easily centered between vertical supports
- Vertical posts could be manufactured inexpensively

Cons

- Large risk of binding if the 4 independent rods are not in perfect alignment
- More components, generally more complex design
- The effect that a bending moment would have on the vertical movement is unknown

The initial concept for the vertical lift mechanism was intended to have a high strength to weight ratio, however it proved to be too complex. Additionally, with the unknown effect that a bending moment would have on the vertical movement, a simpler design was sought after.

Iteration 2 - Single post guide with off center linear actuator (First Revision of Design)

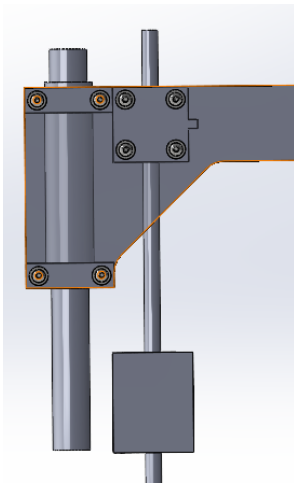


Figure 12: Single post linear guide

Pros

- Very simple design with minimal parts.
- 3/4" diameter metal guide post would easily resist bending moment.

Cons

- Tight tolerance required between guide and sleeve.
- The effect that a bending moment would have on the vertical movement is unknown.

Although simpler in design and higher in strength than the concept, the first design revision was still vulnerable to binding of the slider if the guide post deflected due to loading. Because of this, research was done into off-the-shelf linear movement solutions. This led to the next, and final, design revision.

Iteration 3 - Linear slide with off center linear actuator (Final Revision of Design)

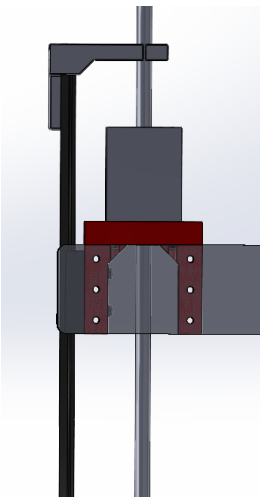


Figure 13: Linear rail and actuator

Pros

- Linear rail and linear actuator can be tested immediately for failure to lift 15 lbs.
- Only the components to integrate actuator and slide rail to the rest of the manipulator required to be designed.
- Linear rail can take large bending moments without binding.

Cons

- Linear rail and actuator are both expensive components.

The final iteration of the vertical lift was the simplest and by far the most refined iteration of the subsystem. Using an off the shelf linear actuator and slide rail provided for a strong design resistant to binding when loaded to its maximum capacity. Additionally the use of the linear actuator allowed for high positioning precision of the boom.

2.2.4 Rotating base design iterations

As with the vertical lift, the design of the rotational base was iterated throughout the design process. Although strength and smooth operation were the major design considerations, the driving force for these iterations was the budget, as discussed in the following iterations.

Iteration 1 - Rotating base with purchased bearing mount

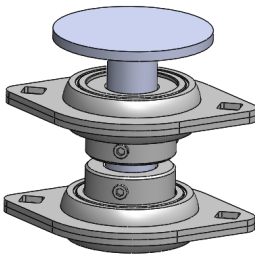


Figure 14: Rotating base using purchased parts

(ring gear, bearing standoffs and stepper motor mount removed for clarity)

Pros

- Simple, inexpensive design.
- Off-the-shelf parts would make for quick assembly once ordered.

Cons

- Bearing misalignment may cause binding of rotating shaft.
- Appears 'unfinished' in design.

Due to the tight budget, the initial design for the rotating base primarily included off-the-shelf parts to form the rotating base. However, when the horizontal positioner was cut from the boom design, manufacturing components out of aluminum became affordable, leading to the next and final iteration of the base.

Iteration 2 - Machined aluminum rotating base

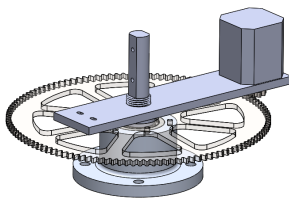


Figure 15 : Final rotating base design

Pros

- Much stronger one piece aluminum design for bearing housing.
- Precision machined aluminum eliminated any play in rotating components that could cause binding when loaded.

Cons

- Aluminum was an expensive material in raw form.

By machining the bearing carrier, stepper mount and rotating shaft out of aluminum, the rotating base design became much stronger and lighter than the original design with purchased steel components. The exact specifics of this will be discussed in section 2.3.

2.3 Final design

As a result of the iterative design process outlined in section 2.2, the final design of the manipulator was finalized and ready to be fully manufactured and assembled. This final design can be seen in figure 16 to the right, and its specifics will be overviewed in the following subsections. In addition to each overall subassembly, the major components of each will be reviewed, including the purpose of that component and the design considerations behind it.

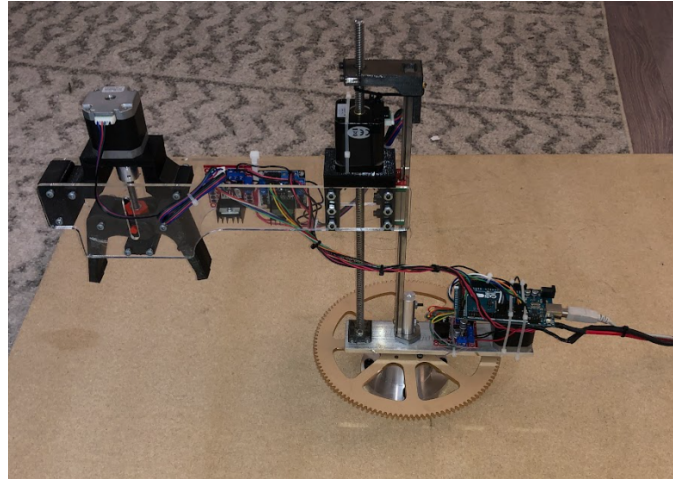


Figure 16 : Final design of manipulator

2.3.1 Gripper sub-assembly

The first subassembly that will be broken down is that of the gripper, shown in (figure 17) on the right. Overall this subassembly is the most complex with all of its moving parts. The final gripper consisted of a stepper mount, shaft coupler, sled, gripper arms, a threaded rod, a bushing for the rod, and a housing for that bushing. These components will be highlighted and explained in the following table. (Table 4)

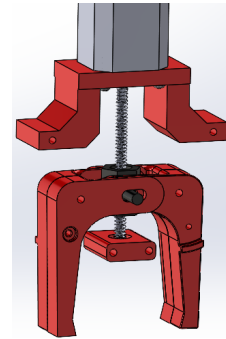
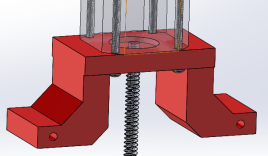
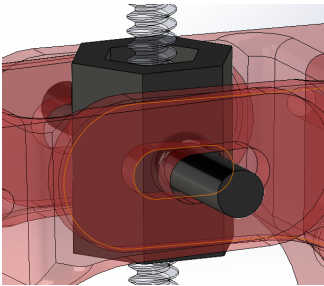
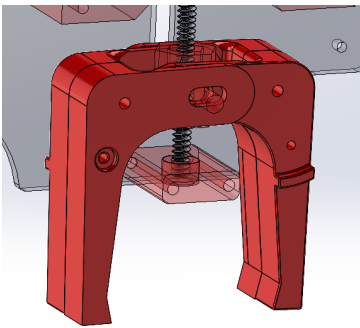
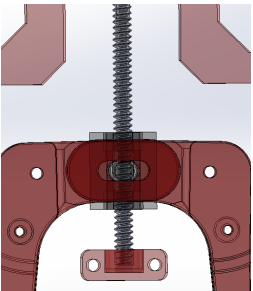
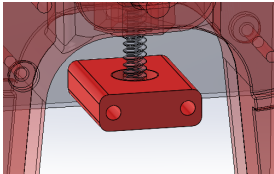
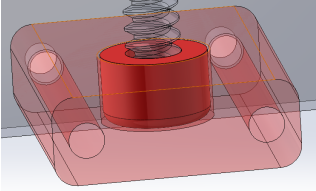


Figure 17 : Final gripper design

Table 4: Gripper subassembly component overview

Component	Description
Stepper mount  Figure 18 : Stepper Mount	This piece, made of 70% infill PLA, supports the stepper motor which drives the gripper. It is bolted with 4 through holes that line up with the M3 threaded holes in the stepper motor. The stepper mount also acts as support for the boom using two bolt holes at the end to keep the tip of the boom rigid.

Component	Description
Coupler  Figure 19: Machined Coupler	This component attaches to the stepper motors drive shaft and translates rotational motion to the threaded rod. Originally made with 20% infill PLA, it was revised and machined of aluminum. The revised design contains a set screw that threads tightly against the flat spot on the steppers drive shaft to prevent the stepper from spinning relative to the coupler. Additionally, the set screw also makes the height of the coupler adjustable relative to the stepper motor shaft. A shear pin is press fit through a hole on the coupler through a matching hole on threaded rod. This prevents the coupler spinning relative to the threaded rod.
Sled  Figure 20: Gripper Sled	The sled, made from 20% infill PLA, remained unchanged since the original prototype. This component is necessary to translate the rotational motion from the stepper to linear motion for actuation of the gripper arms. Two nuts are pressed into either side of the sled which then thread onto the threaded rod. The two pegs that stick out either side go through the slot in the gripper arms and the slot in the chassis. These pegs fit in the chassis to constrain the sled from spinning freely with the stepper motor, forcing the sled to translate linearly up and down the rod. When the sled moves up the threaded rod it pivots the gripper arms forcing the bottom of the gripper arms to close, and vice versa on the way down.
Gripper arms  Figure 21: Gripper Arms	The gripper arms are what physically attaches to the object that is being moved. They are made from 100% infill PLA from the 3D printer. They pivot about a 10-32 bolt that attaches through the chassis which acts as an axle. Each gripper arm is made from two halves (4 total components). This was done because of a geometrical restriction that would have prevented the sled from being installed if they were made from one piece. To keep the two halves of each gripper arm together, a coarse thread screw is installed with a counterbore. These gripper arms are larger than that of the original prototype because the original arms failed under fatigue loading when trying to lift heavy objects repeatedly. At the bottom of the gripper arms there is a small angled outward section. This helps the gripper attach to round objects easier, specifically for dumbbells when testing the strength of the gripper.

Component	Description
Threaded rod  Figure 22: Threaded Rod	The threaded rod in combination with the sled translates rotational motion from the stepper motor to linear motion. This rod, which remained unchanged from the gripper prototype, was cut from a length of ¼”-28 UNF threaded rod found in MESS. A fine thread was chosen in order to attain a higher mechanical advantage. With a higher thread per inch count, this rod would require more revolutions of the stepper motor to actuate the gripper, but would do so with a higher torque output. This allowed the extremely high grip strength needed in order to achieve a high strength to weight ratio.
Bushing housing  Figure 23: Bushing Housing	The bushing housing supports the bottom of the threaded rod to prevent it from pulling itself out of the coupler when under load. It is bolted to the chassis with two 2” long 10-32 bolts. There is a hole that goes half way through the entire piece in the center which contains the threaded rod bushing allowing it, and the threaded rod, to spin relatively easily.
Threaded rod bushing  Figure 24: Threaded rod bushing	The threaded rod bushing screws onto the bottom of the threaded rod and enables it to spin within the threaded rod bushing housing. The only purpose of this piece is to prevent the threads on the rod from cutting into the housing and pulling up on it, which would increase rotational friction.

2.3.2 Boom sub-assembly

The second subassembly that will be reviewed is that of the boom, shown in figure 25 on the right. Overall this subassembly was the simplest as everything on the boom moved linearly up and down together. The final boom consisted of the chassis, a linear actuator to slide rail mount, as well as the gripper subassembly. These components will be highlighted and explained in the following table. (Table 5)

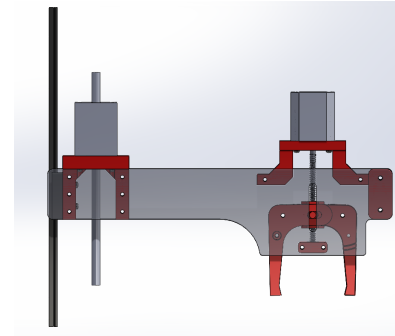
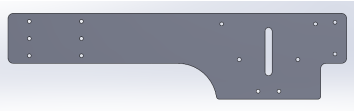
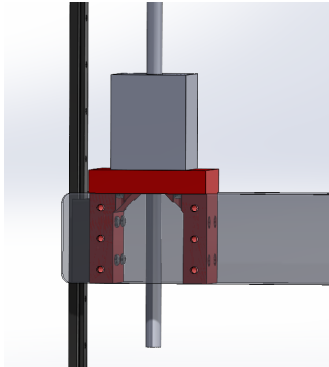
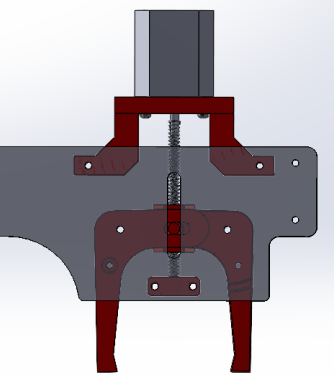
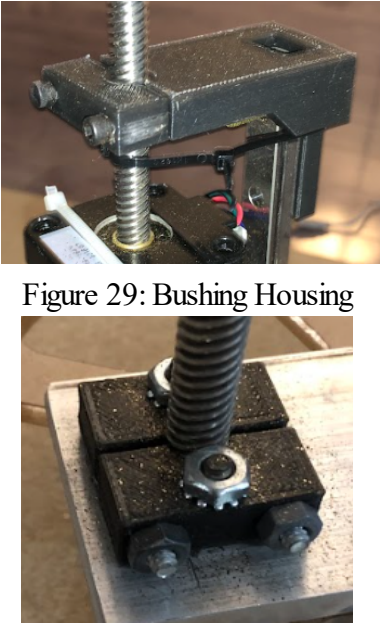


Figure 25: Final boom design

Table 5: Boom subassembly component overview

Component	Description
Chassis  Figure 26: Chassis Panel	Physically supports all components in the boom sub-assembly, this includes the integrated gripper, linear actuator and linear rail interface. Laser cut out of 4.5 mm thick acrylic. Acrylic proved to be sufficiently strong from previous tests and is extremely light compared to the rest of the design.
Linear actuator to slide rail mount (red)  Figure 27: Linear actuator to slide rail mount	Mechanically integrates the chassis to the linear actuator (grey) and to the linear rail (black). This component takes a large moment created by the cantilevered weight at the end of the boom. Thus, it was made of 100% infill PLA. It was mounted to the chassis with 6 evenly spaced bolts to reduce the chance of bolts shearing out through plastic. This increases this component's resistance to bending caused by cantilevered weight at end of boom. Also, it was mounted with 4 bolts and washers into the linear rail and was designed with a contact fit to reduce the chance of the bolts pulling out when the component was subjected to a large moment.

Component	Description
Integrated gripper  Figure 28: Integrated Gripper	This gripper mechanism is a slightly modified version of our original gripper prototype which had proved to be a very successful design. Many cycles of testing lead to the gripper arm failing which lead us to increase the size of the gripper arms as well as increasing print infill from 20% to 100%. The original gripper prototype had a smaller chassis that only enclosed the grippers components. To increase the strength and reduce the number of parts that make up the boom, the gripper was integrated into the chassis itself.
Linear actuator clamps  Figure 29: Bushing Housing Figure 30: Integrated Gripper	These clamps prevent the linear actuator threaded rod from rotating when the linear actuator is subject to vertical load. Because of equal and opposite reaction forces when the linear actuator attempts to spin itself and translate linearly it also attempts to spin the rod. Both top and bottom clamps are printed using 100% infill PLA. They work by “sandwiching” the threaded rod between the two halves of the clamp, using the force of friction to prevent rotation. The bottom clamp is bolted to the rotating plate and the top clamp is bolted to the top two holes of the linear rail. Displayed in figure 29 on the left, a zip tie was used for additional support, this was added during testing of max weight when we saw the linear actuator pulling away from the linear rail.

2.3.3 Rotating base sub-assembly

The last subassembly that will be reviewed is that of the rotating base, shown in figure 31 on the right. Overall this subassembly had the tightest tolerances as fitment of the shaft, bearings and bearing housing had to be perfect to ensure the system had no play that could reduce the accuracy of object placement. The final rotating base assembly consisted of the bearing housing, the rotating shaft, the mounting plate, and both the ring and pinion gears. These components will be highlighted and explained in the following table. (Table 6)

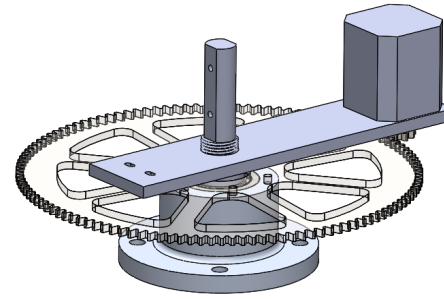
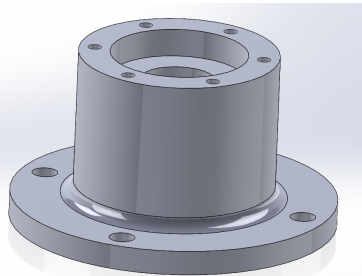
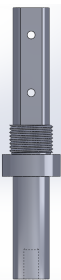
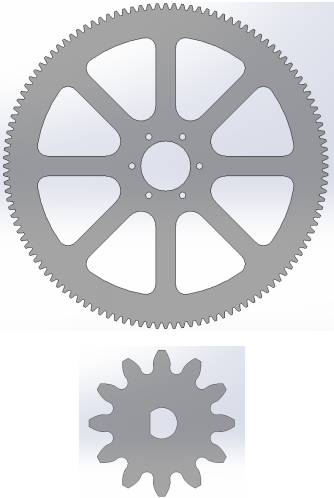


Figure 31 : Final rotating base design

Table 6: Boom subassembly component overview

Component	Description
Bearing housing  Figure 32: Bearing Housing	The bearing housing was necessary to provide a means for mounting the manipulator to the base structure, as well as providing smooth and resistance free rotation. It housed two off-the-shelf ball bearings which then support the rotating shaft. Mounting directly to the base structure with four 5/16” bolts, the bearing housing provided a sturdy base for the entire assembly. Additionally, the ring gear mounted directly to the top surface of the bearing carrier with six 8-32 machine screws. Because the bearing housing must take the entire moment created by the boom with no deflection, it was oversized and machined from aluminum which has a relatively high Young’s Modulus. Additionally, because it is softer than steel, the aluminum parts were relatively easier to machine.
Rotating shaft  Figure 33: Rotating Shaft	The rotating shaft’s purpose was to hold up the vertical lift and boom while allowing them to rotate without resistance no matter what the load was on the boom. To accomplish this, the portion of the shaft in the bearings was $\frac{3}{4}$” in diameter to reduce the chances of deflection. To maximize the accuracy of the manipulator, this shaft was also carefully machined so that there was no play between it and the bearings. Additionally, the shaft supported the mounting plate for the stepper motor by pinching it against the shoulder on the shaft with a nut. To mount the vertical slide rail, through-bolts were used on the flat portion of the shaft. The shaft was made out of aluminum due to its strength, minimized weight and ease of machining.

Component	Description
Mounting plate  Figure 34: Mounting Plate	The mounting plate attached to the rotating shaft with a nut and held the stepper that provided the rotational control of the manipulator. Additionally, this plate supported the base of the linear actuator's threaded rod. Because of this, $\frac{1}{4}$" thick AISI 6061 aluminum plate was chosen over 3d printing because this plate had to carry the entire load held by the boom. Another reason why aluminum was chosen is due to the fact that the bolt holes for the stepper mount could be machined in it precisely enough in order to get the gears to mesh properly.
Gears  Figure 35: Ring and Pinion Gears	Gears were used to convert the fast rotation of the stepper motor into a slow, smooth rotation of the manipulator. A gear reduction of 10 to 1 was desired as that would drop the manipulator rotational speed to 3 rpm when we ran the stepper at 30 rpm. This would allow us to turn the manipulator 180° in 10 seconds. At slower speeds, the manipulator's rotational position would be more accurately controlled as it rotates smoother. Because the gears were to be manufactured out of 4.5 mm acrylic, the gear calculations were done such that the tooth size remained as large as possible while attaining the desired ratio and keeping the gear under 10 inches in diameter so that the teeth wouldn't break and the manipulator would remain compact. To minimize the number of wires that would be twisting around the base, it was decided that the stepper motor and pinion gear would be mounted to the rotating portion of the manipulator with the mount plate and the ring gear would be fixed. By doing this, all of the electronics could then be mounted to the rotating part of the manipulator with only power leads and a usb cable required to hang off the manipulator.

3.0 Design Verification

3.1 Final Design Verification

Since most of the engineering design specifications are related to the performance of the system, the verification strategies were decided through testing the system under several different situations and with different goals in mind depending on each respective specification.

Based on the design specifications that were stated previously, there were certain parts of the system that had to be tested or verified before finalizing them. While considering the strength to weight ratio specification, each component needed to be sufficient for lifting heavy weights in relation to the total weight. This was mostly accomplished by prototyping parts and adding different loads to make sure that they would be able to carry enough weight without deflecting or yielding.

For the precision specification, it was decided to use stepper motors for the system as they are precise and can be easily coded to move exactly as desired. In addition to precision, the system must be consistent over several repetitions, this was verified by making sure we had well meshed gears to avoid any skipping or grinding and minimal slack on the system.

3.2 Detailed Verification Strategy

3.2.1 3D Benchy Test

The test begins with placing a 3D Benchy on the platform which the system is bolted to. Then, by running a set program, the system must pick up, translate and place the benchy in its designated drop location. According to project requirements, the Benchy should be lifted above 2 inch walls.

In the test, the benchy was lifted well over 2 inches and translated precisely to the desired location. Therefore, the system successfully completed this specification. This can be seen in the test video submitted with the final report.

3.2.2 Strength to Weight Ratio

The minimum required strength to weight ratio was at least 1:1. This meant that the system must be able to grip, pick up, translate, and place an object of equal weight (approximately 6.8 pounds)(Figure 36). The system included the weight of the rotating base, boom, gripper, and controls components (does not include the weight of the wooden platform which the rotating base is bolted to).

To test this, the system was incrementally given heavier objects to lift, starting with 5 lbs and increasing by 2.5 lbs each time. The system failed to carry the load at 17.5 lbs but was able to successfully lift the

“15 lbs” object. After measuring the exact weight of the object, it was found to be 16.6 lbs(Figure 37). Therefore the maximum strength to weight ratio was 2.44:1 which greatly exceeded the initial goal. (Refer to Test Video)

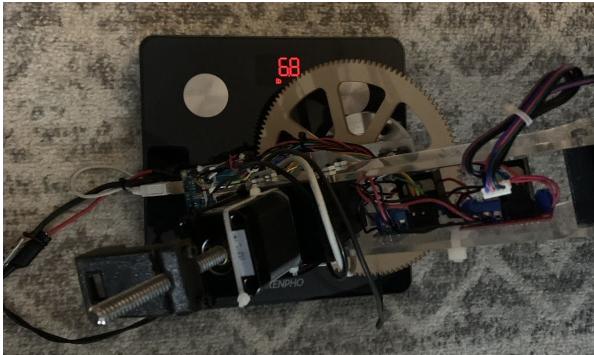


Figure 36: Weight of System



Figure 37: Weight of Object

3.2.3 Precision

The system must be able to repeatedly pick and place an object with an error less than 2% of its radius. In this test, a 3” diameter hockey puck was used as a radially symmetric object.

A target was printed out with several circles at varying diameters, starting with 3 inches and expanding to 3.0625, 3.125, 3.25, and 3.375 inches respectively(Figure 38). A code was run to pick and place a puck from the pickup location to the target and each repetition was recorded. It was seen that the puck was always within or on the 3” diameter line. Therefore the error of the system was well below 2% of its radius. (Refer to Test Video)

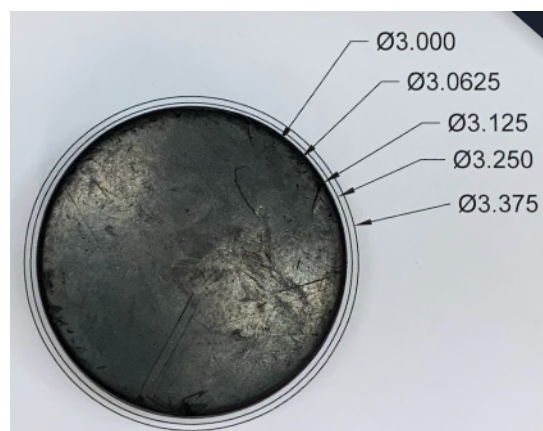


Figure 38: Target for Precision Test

3.2.4 Consistency

The system should have repeatable consistency over multiple uses. This means that after several repetitions, the system should not deviate from the intended pick up and drop off locations.

To test this, six pucks were stacked together at one location and a code was run to stack these pucks into a pyramid shape and then return the pucks back into a stack in the initial location. The system was able to successfully create a pyramid and return the pucks to a stack. (Refer to Test Video)

Original Stack

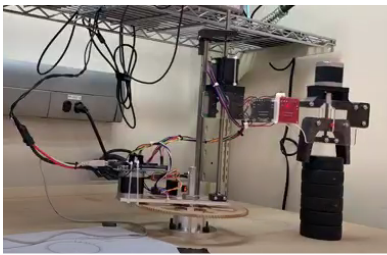


Figure 39

Pyramid

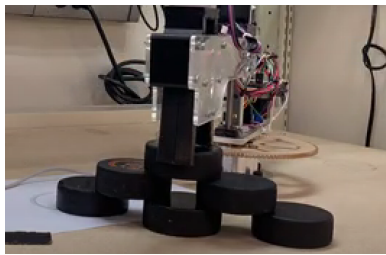


Figure 40

Final Stack

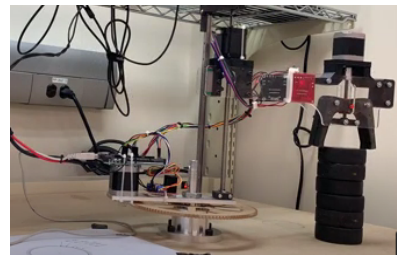


Figure 41

4.0 Project Management

Project management mainly included outlining progress and objectives in weekly meetings. At least once a week the group would meet in-person to discuss current individual progress, as well as future deadlines and goals. These goals and deadlines were further organised in a Gantt chart which outlined the entire project from start to finish. By clearly defining the deadlines for the completion of critical objectives such as the integration of the rotational/vertical systems to the boom/gripper assembly and the final controls integration well in advance, the Gantt chart provided each team member a clear timeline which had to correspond with their respective progress. It also ensured all aspects of the project's completion were at least considered in advance which minimized the possibility of conflicts.

4.1 Time and Budget Changes

For the majority of the project, the initial Gantt chart projections had been quite accurate given a consistent work effort. Towards the tail end of the project the group began to miss deadlines such as the integration of the boom/gripper and the rotating base assembly. This was a result of a delay in sourcing/manufacturing various parts as well as some small final design adjustments. However, because the Gantt chart had previously accounted for an additional week of margin, this was not an issue and the manipulator was fully completed before the deadline.

Similarly, the group organised its finances in an excel sheet which was updated frequently to ensure the maximum budget was not exceeded. Additionally, price estimates of major components such as the linear slide, stepper motors, arduino bridges etc. were added to the budget early. This provided the group with a clear view into the finances which were still available at any time. This resulted in a final budget of \$265.57, \$34.43 less than the \$300 maximum budget.

4.2 Risk Management

From the initial stages of conceptual design, and throughout the remainder of the project, several potential risks were identified. The first of these risks was the possibility of physical incompatibility between sub-assemblies and various parts. Because the mechanical design work was split between multiple team members, along with the design software of choice being solidworks which does not have a "shared work" feature, interference was always a perceived risk. To avoid this all parts and sub-assemblies were updated in the master assembly after any new changes were made. Once the final design work had been completed, the master assembly was inspected again to ensure full functionality before manufacturing began. This resulted in all the manufactured parts working as intended without the need to rebuild any, saving time and budget.

The next risk was identified as the possibility of the various stepper motors lacking the power to perform their intended tasks. Because one of the main design parameters was a strength to weight ratio

of 1:1 or higher, the stepper motors would be the limiting factor in achieving this. Thus, to ensure the lifting capacity of the manipulator was sufficient, each stepper motor was tested individually with approximate loads. Through this testing the stepper responsible for outputting the most force, the linear actuator, could lift well over 10 pounds which indicated the final assembled manipulator would achieve its 1:1 ratio safely.

The final identified risk resulted from the possibility of fatigue failure amongst the structural components. Many of these components were made from either acrylic (boom, gears) or 3D printed PLA (gripper components, brackets, etc.) which are generally lacking in terms of endurance strength when compared to materials such as steel. This along with various large stress concentration factors meant failure was an important consideration. To mitigate the possibility of such failure, fillets and circular holes were predominately used instead of perpendicular edges in the design. Also during testing, crack propagation was closely monitored to ensure no failure.

5.0 Conclusions

5.1 Verification Questions

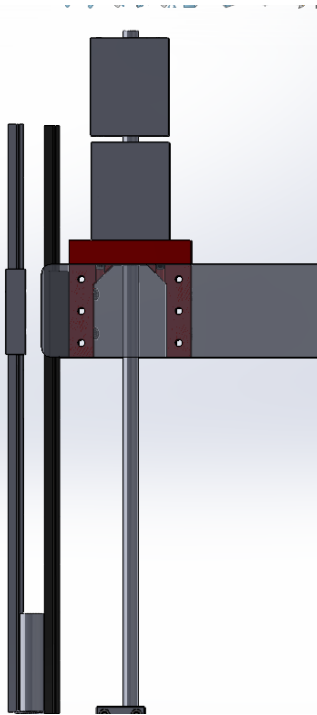
5.1.1 Design Implementation

The specification which best demonstrates the completion of our intended design objective was the 2.44:1 strength to weight ratio. Given that our initial objective was to achieve a strength to weight ratio of at least 1:1, the final result clearly shows our design exceeded expectations.

5.1.2 Specifications

To improve the verification of our specifications given no time constraint, some changes could be made. The first of these changes would be to use smaller denomination weights when testing the maximum lifting capacity. This would allow us to determine a more precise strength to weight ratio. Also, the addition of a set pick up holder would have eliminated any user error in measuring the manipulator's precision.

5.2 Limitations/Future Improvements



Given a larger time frame and slightly larger budget, we could have improved on our primary objectives without any significant changes. To improve our robots strength to weight ratio, we could have added in another linear actuator to the same threaded rod the current linear actuator mounted to. Our current manipulators lifting abilities are limited by the linear actuators ability to resist vertical load. This was qualitatively determined by the grinding noises the linear actuator made when attempting to lift above the maximum capacity. By installing another linear actuator on top of the current one, we theoretically could get twice the amount of vertical lifting force.

The next limiting factor that was observed when increasing lifting weight was the strength of our linear rail. When the manipulator lifted its max weight, we could see slight deflection in the linear rail. To improve this we would bolt another linear slide to the other side of the rotating shaft and mount the boom to two independent linear slides (Figure 42).

Figure 42: Theoretical Vertical
Slide improvements

To improve on the manipulator's accuracy, the next step would be achieving smoother rotational movement. Currently, our rotational stepper motor is set to full speed with no time for acceleration. This results in jerky motion at the beginning of rotation, especially when lifting a heavy weight. To make this improvement we would implement an arduino function that slowly brings the stepper motor up to its desired rotating speed as well as slowly decelerating when coming to a stop. Although this did not impact our current precision objectives, if in the future we needed to lift more weight with a higher precision this would need to be implemented into the software.

5.3 Final Remarks

In conclusion, the objective of this report was to outline the design of a pick and place manipulator from initial conceptualization to final verification. The manipulator was required to grab a 3D benchy, lift it at least 2 inches vertically, move it to a designated drop-off location and finally place it. Additionally, the manipulator required a strength to weight ratio of at least 1:1, while costing a maximum of \$300.

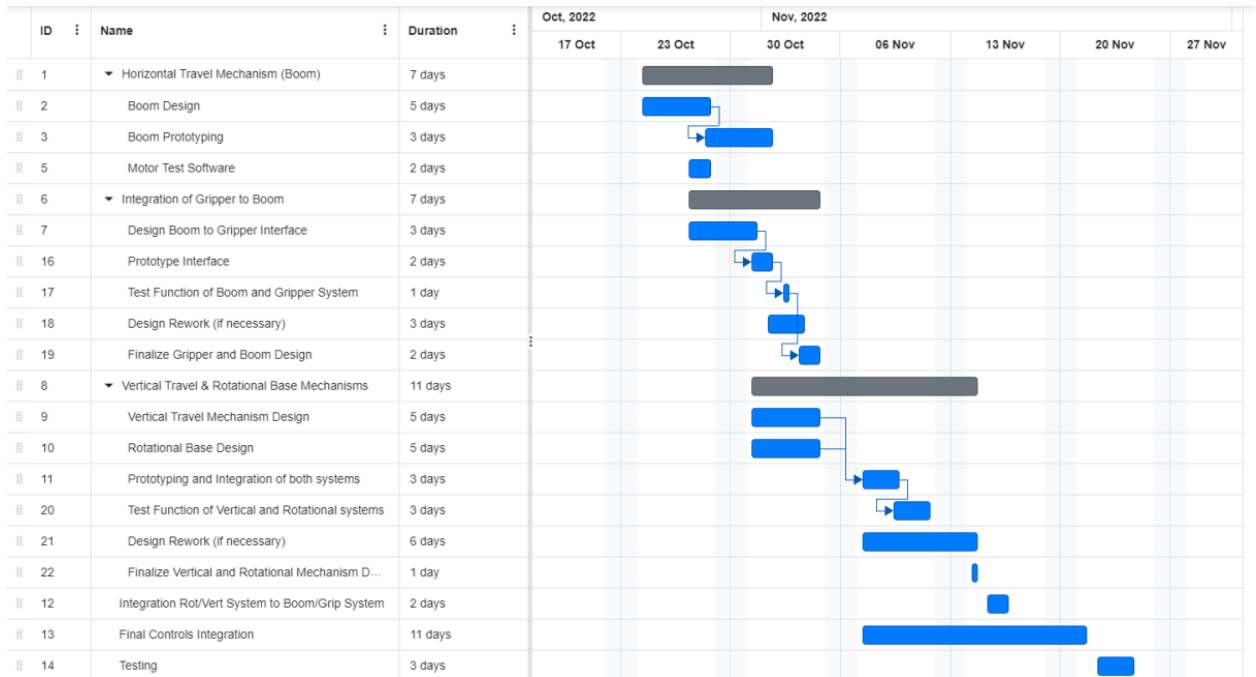
To achieve these specifications, the manipulator would be split into two main sub-assemblies: the rotating base/vertical slide and the horizontal boom/gripper. The rotating base consisted of an aluminum bearing housing with two press-fit bearings which would rotate with an aluminum shaft bolted to a vertical slide and mounting plate. The housing was also attached to a large stationary gear which would allow the entire assembly to rotate. The horizontal boom consisted of a 3D printed gripper sandwiched between two laser cut acrylic slabs. It was attached to the vertical slide via a 3D printed mount. Three stepper motors provided the required motion.

To verify the initial design requirements, three tests were conducted. The first test was a test involving picking and placing the 3D benchy. This test was pass or fail in which the manipulator passed. The next test was the strength to weight ratio test which the manipulator exceeded, achieving a final strength to weight ratio of 2.44:1. The final test was an accuracy test in which the manipulator picked and placed a hockey puck with a positional error well under 2% of the hockey puck's radius.

These tests provided data necessary to prove all of the intended requirements, with a final cost of \$265.57.

6.0 Appendices

1. Gantt Chart:



2. Bill of Materials:

BOM					
Subassembly	Component	Source	Qty	Individual Cost	Total Cost
Rotating Base	Bearing Housing	McMaster-Carr	1	\$11.75	\$11.75
	Vertical Shaft	Manufactured	1	\$2.14	\$2.14
	Stepper Mount Plate	Manufactured	1	\$2.59	\$2.59
	Ring Gear	Manufactured	1	\$2.04	\$2.04
	Pinion Gear	Manufactured	2	\$0.03	\$0.06
	Bearing (MC# 60355K507)	McMaster-Carr	2	\$8.53	\$17.06
	M3 x 10mm BHSS	McMaster-Carr	2	\$0.70	\$1.40
	M4 x 25mm SHCS	McMaster-Carr	2	\$0.20	\$0.40
	M4 x 25mm FHSS	McMaster-Carr	2	\$0.11	\$0.22
	8 - 32 x 0.5" SHCS	McMaster-Carr	6	\$0.14	\$0.84
	5/16" - 18 x 1" HHCS	McMaster-Carr	1	\$0.28	\$0.28
	5/16" - 18 x 1.75" SHCS	McMaster-Carr	4	\$0.39	\$1.56
	5/16" Washer	McMaster-Carr	5	\$0.25	\$1.25
	7/16" Washer	McMaster-Carr	4	\$0.60	\$2.40
	M4 Nut	McMaster-Carr	4	\$0.03	\$0.12
	5/16" - 18 Nut	McMaster-Carr	4	\$0.13	\$0.52
	3/4" - 16 Nut	McMaster-Carr	1	\$0.90	\$0.90
Vertical Assembly	Linear Actuator Screw Clamp - Top	Manufactured	1	\$1.60	\$1.60
	Linear Actuator Screw Clamp - Bottom	Manufactured	1	\$0.80	\$0.80
	Slide Rail	MESS	1	\$40.00	\$40.00
	M4 x 25mm SHCS	McMaster-Carr	1	\$0.20	\$0.20
	6 - 32 x 1.25" BHCS	McMaster-Carr	2	\$0.10	\$0.20
	6 - 32 x 2" SHCS	McMaster-Carr	2	\$0.20	\$0.40
	6 - 32 Nut	McMaster-Carr	2	\$0.01	\$0.02

BOM					
Subassembly	Component	Source	Qty	Individual Cost	Total Cost
Horizontal Boom and Gripper	Boom Panel	Manufactured	2	\$2.07	\$4.14
	Linear Acuator Mount	Manufactured	1	\$4.20	\$4.20
	Gripper Stepper Mount	Manufactured	1	\$3.80	\$3.80
	Boom End Spacer	Manufactured	1	\$1.80	\$1.80
	Gripper Claw - RH Side	Manufactured	2	\$1.00	\$2.00
	Gripper Claw - LH Side	Manufactured	2	\$1.00	\$2.00
	Slider	Manufactured	1	\$1.00	\$1.00
	Threaded Rod End Support	Manufactured	1	\$1.00	\$1.00
	Threaded Rod Bushing	Manufactured	1	\$0.20	\$0.20
	Stepper Motor Coupler	Manufactured	1	\$0.35	\$0.35
	M4 x 25mm SHCS	McMaster-Carr	4	\$0.20	\$0.80
	6 - 32 x 2" SHCS	McMaster-Carr	14	\$0.20	\$2.80
	8 - 32 x 0.5" SHCS	McMaster-Carr	1	\$0.14	\$0.14
	1/4" - 28 x 2.5" Threaded Rod	McMaster-Carr	1	\$6.80	\$6.80
	6 - 32 Nut	McMaster-Carr	18	\$0.01	\$0.18
	1/4" - 28 Nut	McMaster-Carr	2	\$0.08	\$0.16
Controls	Dual H Bridge L298N	Amazon	3	\$11.00	\$33.00
	Stepper Motor NEMA17	Amazon	2	\$20.00	\$40.00
	Linear Actuator Motor NEMA14	MESS	1	\$40.00	\$40.00
	Arduino MEGA 2560	Amazon	1	\$30.00	\$30.00
	Jumper Cable	MESS	12	\$0.03	\$0.36
	Main Power Wires	MESS	2	\$0.05	\$0.10
	Zipties	Amazon	30	\$2.00	\$2.00
TOTAL					\$265.57

3. Code for Manual Control:

```
#include <Stepper.h>
```

```
//initialize variables and motors
```

```
int stepsPerRevolution1 = 200;
```

```
int stepsPerRevolution2 = 200;
```

```
int stepsPerRevolution3 = 200;
```

```
String incomingByte;
```

```
Stepper myStepper1(stepsPerRevolution1, 8, 9, 10, 11);
```

```
Stepper myStepper2(stepsPerRevolution2, 2, 3, 4, 5);
```

```
Stepper myStepper3(stepsPerRevolution3, 22, 24, 26, 28);
```

```
int motorSpeed = 60;
```

```
bool runMotor1 = false;
```

```
bool runMotor2 = false;
```

```
bool runMotor3 = false;
```

```
void setup() {
```

```
  myStepper1.setSpeed(60);
```

```
  myStepper2.setSpeed(60);
```

```
  myStepper3.setSpeed(60);
```

```

Serial.begin(9600);
Serial.setTimeout(10);

}

void loop() {
  if (Serial.available () > 0) {
    incomingByte = Serial.readString();
    Serial.print("I received: ");
    Serial.println(incomingByte);
    stepsPerRevolution1 = 0;
    stepsPerRevolution2 = 0;
    stepsPerRevolution3 = 0;
  }
  switch (incomingByte[0]){

    case '0': //motor 1 stop
      runMotor1 = false;
      break;

    case '1': //motor 1 clockwise
      stepsPerRevolution1 = 10;
      runMotor1 = true;
      break;

    case '2': //motor 1 counter-clockwise
      stepsPerRevolution1 = -10;
      runMotor1 = true;
      break;

    case '3': //motor 2 clockwise
      stepsPerRevolution2 = 10;
      runMotor2 = true;
      break;

    case '4': //motor 3 counter-clockwise
      stepsPerRevolution2 = -10;
      runMotor2 = true;
      break;
  }
}

```

```

case '9': //motor 2 stop
    runMotor2 = false;
    break;

case '5': //motor 3 clockwise
    stepsPerRevolution3 = 10;
    runMotor3 = true;
    break;

case '6': //motor 3 counter-clockwise
    stepsPerRevolution3 = -10;
    runMotor3 = true;
    break;

case '8': //motor 3 stop
    runMotor3 = false;
    break;
}

//code to set motor speed and steps for each motor
if(runMotor1 == true){
    myStepper1.setSpeed(motorSpeed);
    myStepper1.step(stepsPerRevolution1);
}
if(runMotor2 == true){
    myStepper2.setSpeed(motorSpeed);
    myStepper2.step(stepsPerRevolution2);
}
if(runMotor3 == true){
    myStepper3.setSpeed(motorSpeed);
    myStepper3.step(stepsPerRevolution3);
}
}

```

4. Puck pyramid code

```

#include <Stepper.h>

int stepsPerRevolution = 200;

int LAvertical = 17*200;

```

```
int GPgrab = 4*200;
int RBrotate= 4*200;
int integer2=0;
int integer=0;
int reps=5;
Stepper LA(stepsPerRevolution, 8, 9, 10, 11);
Stepper RB(stepsPerRevolution, 2, 3, 4, 5);
Stepper GP(stepsPerRevolution, 22, 24, 26, 28);
```

```
void setup()
{
  LA.setSpeed(120);
  RB.setSpeed(60);
  GP.setSpeed(60);
  Serial.begin(9600);
  // Serial.setTimeout(10); what does this do

}
```

```
void loop()
{

  while (Serial.available () == 0)
  {}
```

```
integer=0;
integer2=0;
```

```
while (integer<= reps)
{
```

```
  LA.step(-LAvertical+((5-integer)*620)); //move down from origin
  delay(500);
  turnOff();
```

```
  GP.step(-GPgrab); // grab object
  delay(500);
```

```

turnOff();
  LA.step(LAvertical-((5-integer)*620)); //move up with object
  delay(500);
turnOff();
if (integer<=2)
{
  RB.step(RBrotate-(integer*130)); //rotate to drop off
  delay(500);
turnOff();
  LA.step(-LAvertical); //move down to target
  delay(500);
turnOff();
  GP.step(GPgrab); // release object
  delay(500);
turnOff();
  LA.step(LAvertical); //move back up
  delay(500);
turnOff();
  RB.step(-RBrotate+(integer*130)); //rotate back to origin
  delay(500);
turnOff();
}

if( integer>2 && integer<=4)
{
  RB.step((RBrotate-65)-((integer-3)*130)); //rotate to drop off
  delay(500);
turnOff();
  LA.step(-LAvertical+620); //move down to target
  delay(500);
turnOff();
  GP.step(GPgrab); // release object
  delay(500);
turnOff();
  LA.step(LAvertical-620); //move back up
  delay(500);
turnOff();
  RB.step((-RBrotate+65)+((integer-3)*130)); //rotate back to origin
  delay(500);

```

```

turnOff();
}

if (integer==5)
{
  RB.step((RBrotate-130)-((integer-5)*130)); //rotate to drop off
  delay(500);
  turnOff();
  LA.step(-LAvertical+(2*620)); //move down to target
  delay(500);
  turnOff();
  GP.step(GPgrab); // release object
  delay(500);
  turnOff();
  LA.step(LAvertical-(2*620)); //move back up
  delay(500);
  turnOff();

}
integer=integer+1;

}

while (integer2<=reps)
{

  if (integer2==0)
  {
    Serial.println("test");
    LA.step(-LAvertical+(2*620)); //move down for first puck
    delay(500);
    turnOff();
    GP.step(-GPgrab); // grab object
    delay(500);
    turnOff();
    LA.step(LAvertical-(2*620)); //move up with puck
    delay(500);
    turnOff();
    RB.step((-RBrotate+130)); //rotate back to origin with puck

```

```

delay(500);
turnOff();

}
if (integer2==1 || integer2==2)
{
  RB.step((RBrotate-65)-((integer2-1)*130)); //rotate to 2nd row pucks
  delay(500);
  turnOff();
  LA.step(-LAvertical+620); //move down for 2nd row pucks
  delay(500);
  turnOff();
  GP.step(-GPgrab); // grab object
  delay(500);
  turnOff();
  LA.step(LAvertical-620); //move up with puck
  delay(500);
  turnOff();
  RB.step((-RBrotate+65)+((integer2-1)*130)); //rotate back to origin with puck
  delay(500);
  turnOff();

}
if (integer2==3 || integer2==4 || integer2==5)
{
  RB.step((RBrotate)-((integer2-3)*130)); //rotate to 3rd row pucks
  delay(500);
  turnOff();
  LA.step(-LAvertical); //move down for bottom row
  delay(500);
  turnOff();
  GP.step(-GPgrab); // grab object
  delay(500);
  turnOff();
  LA.step(LAvertical); //move up with puck
  delay(500);
  turnOff();
  RB.step((-RBrotate)+((integer2-3)*130)); //rotate back to origin with puck
  delay(500);
  turnOff();
}

```

```

}

Serial.println ("tesst2");

LA.step(-LAvertical+((integer2)*620)); //move down from origin with puck
delay(500);
turnOff();
GP.step(GPgrab); // Release puck on stack
delay(500);
turnOff();
LA.step(LAvertical-((integer2)*620)); //move up from origin
delay(500);
turnOff();

integer2 = integer2+1;
}
}
void turnOff()
{
  digitalWrite (8 , LOW);
  digitalWrite (9 , LOW);
  digitalWrite (10 , LOW);
  digitalWrite (11 , LOW);
  digitalWrite (2 , LOW);
  digitalWrite (3 , LOW);
  digitalWrite (4 , LOW);
  digitalWrite (5 , LOW);
  digitalWrite (22 , LOW);
  digitalWrite (24 , LOW);
  digitalWrite (26 , LOW);
  digitalWrite (28 , LOW);

}

```

5. Percicely place 30 pucks in same place code

```
#include <Stepper.h>
```

```
int stepsPerRevolution = 200;
```

```
int LAvertical = 17*200;
```

```
int GPgrab = 4*200;
```

```
int RBrotate = 5*200;
```

```
int integer=0;
```

```
int reps=35;
```

```
Stepper LA(stepsPerRevolution, 8, 9, 10, 11);
```

```
Stepper RB(stepsPerRevolution, 2, 3, 4, 5);
```

```
Stepper GP(stepsPerRevolution, 22, 24, 26, 28);
```

```
void setup()
```

```
{
```

```
  LA.setSpeed(90);
```

```
  RB.setSpeed(30);
```

```
  GP.setSpeed(60);
```

```
  Serial.begin(9600);
```

```
  // Serial.setTimeout(10); what does this do
```

```
}
```

```
void loop()
```

```
{
```

```
  while (Serial.available () == 0)
```

```
  {}
```

```
  while (integer< reps)
```

```
  {
```

```
    integer= integer + 1;
```

```
    LA.step(-LAvertical); //move down from origin
```

```
    delay(500);
```

```
    digitalWrite (8 , LOW);
```

```
    digitalWrite (9 , LOW);
```

```
    digitalWrite (10 , LOW);
```

```
    digitalWrite (11 , LOW);
```

```
    digitalWrite (2 , LOW);
```

```
    digitalWrite (3 , LOW);
```

```
    digitalWrite (4 , LOW);
```

```
digitalWrite (5 , LOW);  
digitalWrite (22 , LOW);  
digitalWrite (24 , LOW);  
digitalWrite (26 , LOW);  
digitalWrite (28 , LOW);
```

```
GP.step(-GPgrab); // grab object  
delay(500);
```

```
digitalWrite (8 , LOW);  
digitalWrite (9 , LOW);  
digitalWrite (10 , LOW);  
digitalWrite (11 , LOW);  
digitalWrite (2 , LOW);  
digitalWrite (3 , LOW);  
digitalWrite (4 , LOW);  
digitalWrite (5 , LOW);  
digitalWrite (22 , LOW);  
digitalWrite (24 , LOW);  
digitalWrite (26 , LOW);  
digitalWrite (28 , LOW);
```

```
LA.step(LAvertical); //move up with object  
delay(500);
```

```
digitalWrite (8 , LOW);  
digitalWrite (9 , LOW);  
digitalWrite (10 , LOW);  
digitalWrite (11 , LOW);  
digitalWrite (2 , LOW);  
digitalWrite (3 , LOW);  
digitalWrite (4 , LOW);  
digitalWrite (5 , LOW);  
digitalWrite (22 , LOW);  
digitalWrite (24 , LOW);  
digitalWrite (26 , LOW);  
digitalWrite (28 , LOW);
```

```
RB.step(RBrotate); //rotate to drop off  
delay(500);
```

```
digitalWrite (8 , LOW);  
digitalWrite (9 , LOW);  
digitalWrite (10 , LOW);
```

```
digitalWrite (11 , LOW);
digitalWrite (2 , LOW);
digitalWrite (3 , LOW);
digitalWrite (4 , LOW);
digitalWrite (5 , LOW);
digitalWrite (22 , LOW);
digitalWrite (24 , LOW);
digitalWrite (26 , LOW);
digitalWrite (28 , LOW);
  LA.step(-LAvertical); //move down to target
  delay(500);
digitalWrite (8 , LOW);
digitalWrite (9 , LOW);
digitalWrite (10 , LOW);
digitalWrite (11 , LOW);
digitalWrite (2 , LOW);
digitalWrite (3 , LOW);
digitalWrite (4 , LOW);
digitalWrite (5 , LOW);
digitalWrite (22 , LOW);
digitalWrite (24 , LOW);
digitalWrite (26 , LOW);
digitalWrite (28 , LOW);
  GP.step(GPgrab); // release object
  delay(500);
digitalWrite (8 , LOW);
digitalWrite (9 , LOW);
digitalWrite (10 , LOW);
digitalWrite (11 , LOW);
digitalWrite (2 , LOW);
digitalWrite (3 , LOW);
digitalWrite (4 , LOW);
digitalWrite (5 , LOW);
digitalWrite (22 , LOW);
digitalWrite (24 , LOW);
digitalWrite (26 , LOW);
digitalWrite (28 , LOW);
  LA.step(LAvertical); //move back up
  delay(500);
digitalWrite (8 , LOW);
```

```
digitalWrite (9 , LOW);
digitalWrite (10 , LOW);
digitalWrite (11 , LOW);
digitalWrite (2 , LOW);
digitalWrite (3 , LOW);
digitalWrite (4 , LOW);
digitalWrite (5 , LOW);
digitalWrite (22 , LOW);
digitalWrite (24 , LOW);
digitalWrite (26 , LOW);
digitalWrite (28 , LOW);
  RB.step(-RBrotate);//rotate back to origin
  delay(500);
digitalWrite (8 , LOW);
digitalWrite (9 , LOW);
digitalWrite (10 , LOW);
digitalWrite (11 , LOW);
digitalWrite (2 , LOW);
digitalWrite (3 , LOW);
digitalWrite (4 , LOW);
digitalWrite (5 , LOW);
digitalWrite (22 , LOW);
digitalWrite (24 , LOW);
digitalWrite (26 , LOW);
digitalWrite (28 , LOW);
}
}
```